

AI Internship – Day 2 Report

Building a Data Preprocessing Pipeline with Zero Data Leakage

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Objective

The objective of this task was to build a complete machine learning preprocessing pipeline using Scikit-learn that:

- Handles numeric and categorical data separately
 - Fills missing values properly
 - Applies feature scaling
 - Encodes categorical variables
 - Prevents data leakage
 - Saves the final pipeline for reuse
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Dataset Used

- Dataset: California Housing Dataset
 - Target Variable: `median_house_value`
 - Features include:
 - Numeric: longitude, latitude, income, population, etc.
 - Categorical: `ocean_proximity`
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Concepts Learned

1. Feature Separation

The dataset was split into:

- **X (Features)** → input variables
 - **y (Target)** → value to predict (house price)
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2. Train-Test Split

Data was split into training and testing sets using:

- 80% training data
 - 20% testing data
 - Ensures unbiased model evaluation
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3. Column Types Identification

- Numeric columns → handled using scaling
 - Categorical columns → handled using encoding
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4. Pipelines

Two separate pipelines were created:

Numeric Pipeline:

- SimpleImputer (median)
- StandardScaler

Categorical Pipeline:

- SimpleImputer (most frequent)
 - OneHotEncoder
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5. ColumnTransformer

Both pipelines were combined using ColumnTransformer to apply correct transformations to each column type automatically.

6. Data Leakage Prevention

All transformations were:

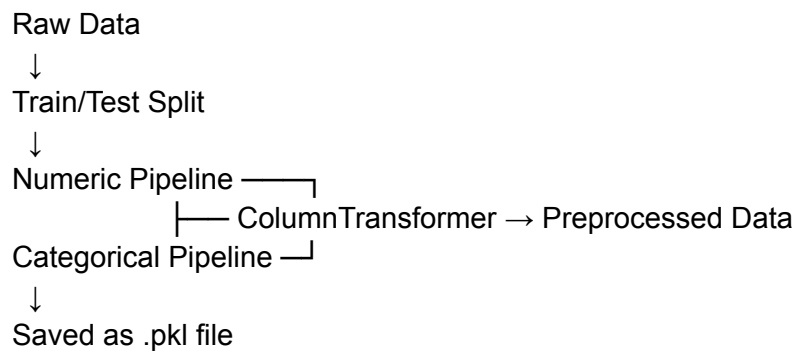
- Fit ONLY on training data
- Applied to test data separately

This ensured no information from test data leaked into training.

7. Pipeline Serialization

The final preprocessing pipeline was saved using joblib for reuse in model training and deployment.

Final Workflow



Tools Used

- Python
 - Pandas
 - NumPy
 - Scikit-learn (Pipeline, ColumnTransformer, SimpleImputer, OneHotEncoder, StandardScaler)
 - Joblib
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Output

- Clean preprocessing pipeline created
 - No data leakage
 - Ready for model training
 - Saved as: `preprocessor.pkl`
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Conclusion

This task helped in understanding how real-world machine learning pipelines are built. The focus was on proper preprocessing, separating data types, and ensuring no leakage between training and testing data. The final pipeline is reusable and production-ready.